

Mouse-tracking in OpenSesame

Goal: familiarise yourself with OpenSesame software and make small modification to the demo so you can collect 1-2 data files to your own experiment next class

- In the next class you'll be collecting some data and analysing it

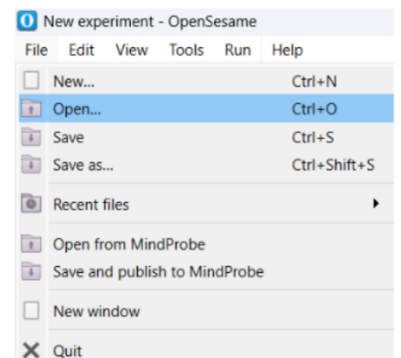
Ideas for a simple modification of the circle-square experiment

When you are done with playing around with the demo experiment, try to come up with an easy (!) modification. For example, you can exchange the shapes with words or images. Or you can add further response options. Or adjust the color contrast.

- Implement these small changes in OpenSesame!

Step 1: Hopefully you managed to successfully install OpenSesame on your computer, if not, find a partner to work together with

Step 1: Open the “circle_square.osexp” experiment from GitHub by going to File -> Open and find the file on your computer



Step 3: Run the experiment but pressing the green arrow



Note: If you would ever like to stop the experiment from full screen mode, press Escape and then finish the trial. Afterwards, a text message asks you to confirm exiting by pressing

Step 4: Understanding the structure of the experiment

The demo experiment has a quite simple, which can be seen in the overview panel

The colourful items are called **sketchpads**, which allow you to present visual information to the participants (text, fixation cross, stimuli, etc.)

Step 4.1: Try changing the text for the instructions

The **block loop** is a table controlling the underlying design of the experiment

Step 4.2: Discuss how the table fits to the experiment you just tried put. What does each column relate to? Now look at the boxes at the top of the page. Try to change something and have a look at the *Preview*.

Trial sequence controls what is happening in each trial (of the underlying items).

Step 4.3: Have a look at it. Why do the values in the “run if” column make sense

Mouse controls what is tracked by the mouse

Step 4.4: Click on the mouse-icon and have a look at the settings:

1. Number of buttons (Number of options you have)
2. Sketchpad (The visual stimuli the participant is presented with)
3. Start coordinates (Where should the mouse be at the beginning of the trial?)
4. Timeout (Is there a time limit, e.g. can you be too slow?)
5. Logger (Storing all information in a .csv file)

Do the names on the settings remind you of anything you have already seen?

Hint 1: Why is [correct_response] in square brackets?

Hint 2: Look in the block_loop

Step 5: Presentation time

In the fixation and stimuli sketchpad you can see in the upper-left corner a box called Duration.

Step 5.1: Click on the *fixation* sketchpad. It shows you an option to adjust the amount of time the fixation point is shown. Make a change!

Step 5.2: Why does it make sense that the presentation time is 0 for stimuli?

Step 6: Stimuli

Stimuli is what the participant is going to see

Step 6.1: Insert a new sketchpad after fixation and draw a square on the sketchpad.

Step 6.2: Click on the square and look at the information about it.

Note: You can also double-click on the item and view the same information in a slightly cryptic code inlet.

Step 6.3: Now go to the other sketchpad stimuli. Move one of the thin lined squares out of the way and then double-click one of the green elements.

Step 6.4: Discuss: What are the differences between the square in the new sketchpad and the green elements in stimuli? Among small other differences, you might recognize

the square brackets again. These mark variables, i.e., references to items defined elsewhere. In this case, the information this refers to is location in the `block_loop` table.

Step 6.5: Delete the `new_sketchpad`.

Step 7: Delete the sketchpad `new_sketchpad` on which you just drew a square

You will see the item is now moved to the Unused items stack.

I recommend that before running your experiment with participants, you remove everything from the Unused items list, so it's not logged.

Final step: Make changes to the experiment to make it your own for data collection next time (e.g. different stimuli, different presentation time etc)

Loading Pictures and Sounds

Pictures are easy to add to sketchpads, like the *stimulus*. Just use the image symbol on the left. Pictures can also have names, so they can also be the targets for the mouse, and there can also be 'show if' statements for pictures.

If we want to add a sound, we will use the *sampler* tool on the left. Here we can add a soundfile (has to be .wav or .ogg), and we can decide for how long it should last with the 'stop after option. If we set the 'duration' to zero, the sound will continue in the background while OpenSesame continues to the next bit – in this way can we play soundfiles while visual stimulus is being shown and while people have to move their mouse. Try them out.

Bugs

- **Problem:** When many tabs are open, you cannot see everything.
Solution: Close all tabs via the View menu.
- **Problem:** Something “weird” is shown on the screen when running the experiment but you can't find that item
Solution: The item might be outside the range of your sketchpad drawing area.
- **Problem:** “Unused items” still somehow pop-up and interfere with the experiment (esp. during data analysis later).
Solution: Delete the list of unused items completely. In fact, it is highly recommended to do this before running any real participants.
- **Problem:** Not visible now, but it seems that setting the start location in the MouseTrap item to something else than [0;0] will later on lead to (not unsolvable but really annoying) problems with the analysis.
Solution: At least for the moment, let's keep it where it is.

- **Problem:** The experiment file can't be opened.
Solution: Try it from within OpenSesame. That seems to work better.